KAPPA FOUNDATION / INVESTMENT COMMITTEE

Initial decision record

USD | Case date 31 December 2022 | Initial submission 22 October 2026

INVESTMENT OBJECTIVE AND POLICY

Meet the fixed education commitments first and protect long-term purchasing power. Invest surplus capital within the fund's capacity for risk. The end-2030 goal is USD 8m in December 2022 purchasing power; it is a goal, not a promised return. Changes require evidence about obligations, resources and assets.

INITIAL CAPITAL ALLOCATION



● Beta 18.46% ● Delta 28.58% ● USMV 52.89% ● Costs 0.08%

HOLDING / USE	UNITS / BASIS	INITIAL USD
Beta | 2024	2,400	2,214,672.00
Delta | 2027	3,478.26087	3,429,808.70
USMV	100% of sleeve	6,346,350.71
Trading costs	5 / 10bp	9,168.59
Total		12,000,000.00

Source: case Exhibit 1; Kappa model; USD, fees separate; percentages rounded.

Beta principal pays $2.4m on 31 Dec 2024. Delta principal and final coupon pay $3.6m on 31 Dec 2027. Four earlier Delta coupons total $486,957 and remain cash at 0%; payment coverage needs no equity sale or reinvestment return.

GROWTH PORTFOLIO DECISION

Recommend 100% USMV, the US minimum-volatility ETF. Use 100% VTI, a broad-US-market ETF, as the frozen investable benchmark and credible rejected alternative.

ANNUAL MEASURE	USMV	VTI	MAX UTILITY A=4
Expected return[1]	7.14%	8.45%	7.19%
Volatility[2]	12.84%	16.31%	12.70%
Utility[3] **| A=4**	3.84%	3.13%	3.96%

Source: Yahoo Finance, Kenneth French; Kappa model; 2015-22 monthly returns, annualised.

Full-universe maximum utility at A=4 (96 months, sample covariance):

MSFT 2.92% | JPM 3.52% | JNJ 7.46% | PG 11.62%
XOM 2.96% | NEE 7.45% | USMV 46.18% | VTI 17.89% | AAPL/CAT 0.00%

This is the highest-scoring long-only mix of the 10 candidates for the stated inputs: CAPM expected returns, sample covariance, weights >=0 and total weight=100%, with no extra caps. It estimates 7.19% return, 12.70% volatility and 3.96% utility, 0.12 percentage point above USMV. The 96-to-60-month check shifts 34.8% of weights one-way (about $2.21m on this sleeve). We choose USMV for simpler policy implementation, not because it is the model maximum.

MANDATE & ASSUMPTIONS

$12m initial; USD 8m real at 2030; CPI 2.5% base, 3.0% education-cost stress.

Forecast RF 4%; market premium 4.5%; forecast alpha 0; provisional A=4.

Coupon cash 0%; 5bp bond / 10bp equity entry cost; no extra position caps.

CANDIDATE UNIVERSE

Technology AAPL, MSFT; financials JPM; health care JNJ; staples PG.

Energy XOM; industrials CAT; utilities NEE; equity ETFs USMV and VTI.

Eight stocks in seven sectors; two ETF mandates. Candidates need not be held.

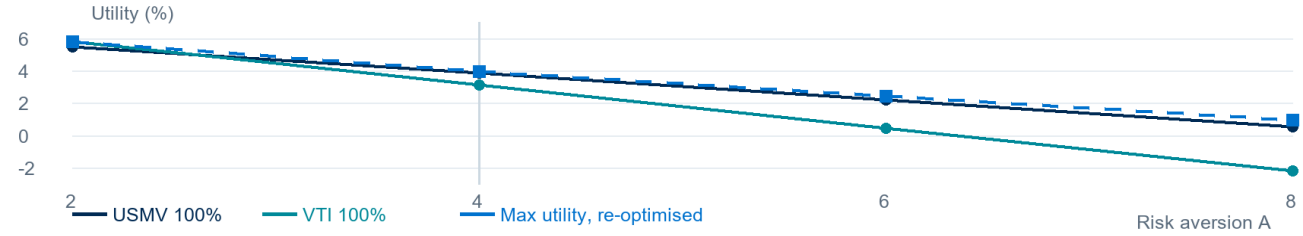
Source: original case, proposed IPS and dated eligibility notes; instruments are long-only, without borrowing, shorting or derivatives.

KAPPA FOUNDATION / INVESTMENT COMMITTEE

Policy, risk and implementation

Growth-sleeve signals | Funding and 2030 goal tracked separately | Proposed policy

RISK PREFERENCE AND REAL GOAL



Source: Kappa analysis from saved Yahoo/French inputs; utility = expected return - 0.5 x A x annual variance.

USMV and VTI stay at 100%. The dashed line re-optimises all 10 candidates at each A and shows the highest estimated score for that risk preference. At A=4 it scores 3.96%; page 1 lists its holdings. Of the fixed ETFs, VTI leads at A=2; USMV leads at A=4, 6 and 8 (crossover A=2.58). A=4 is provisional.

2030 GOAL	CPI BASE	EDUCATION STRESS
Inflation	2.50%	3.00%
Nominal USD	$9.747m	$10.134m
CAGR[4] hurdle	4.84%	5.37%

Source: Kappa funding model; 2022-30; four pre-maturity coupons retained at 0% cash interest.

DOWNSIDE OVERSIGHT

Primary signal: flow-adjusted growth-sleeve total-return drawdown from its running high, including cash; retain the prior peak after changes. Check whole-fund stress and dated-payment coverage separately. These are review triggers, not guaranteed loss limits.

SIGNAL[5]	REQUIRED ACTION	DEADLINE
10%	Validate data, payment coverage and stress	2 business days
15%	Investment Committee decision	5 business days
20%	Notify trustees; pause risk increases	Same / next day
Payment gap	Escalate; restore coverage	Immediate

Source: proposed IPS, 21 Oct 2026; signal is month-end growth-sleeve drawdown, except a payment gap at any time.

No automatic sale. At 15%, record a hold, authorised rebalance or prospective policy proposal. A crisis may defer discretionary action only with written authority, expiry and review; due-payment coverage cannot fall.

ANNUAL GOAL PROTECTION AND POLICY RESPONSE

At each annual review, first confirm remaining bond and cash obligations. Test the 2030 goal of $9.747m nominal at CPI 2.5% ($10.134m at HEPI 3% stress) after a 30% fall in growth assets. If stressed resources still cover the goal, trustees may approve locking only the minimum needed in a permitted, actually priced asset due by 2030; keep excess invested. This is a prospective trustee decision, not automatic profit-taking. If a gap remains, report it; do not raise risk to catch up. For relative underperformance, diagnose exposures, costs and implementation before proposing a change.

FROZEN TEST AND FUTURE REBALANCING

For the test, freeze USMV 100% and VTI 100% of the growth sleeve at 31 Dec 2022. Compare monthly total returns in Jan 2023-Dec 2025. Reset to starting weights after each December return, without updating forecasts. Since each portfolio holds one ETF, this rule normally creates no trade; charge 10bp only when a trade occurs. Later returns test the frozen choice; they do not set the 2022 weights. Future changes require a dated committee review under the IPS.

Estimate Jan 2015-Dec 2022 (96 common months); a 60-month sample check informs the initial recommendation. For the final memo, compare sample with single-index covariance. CAPM forecasts set alpha to zero; FF3 explains historical exposures. No 2023-25 observation enters the 2022 weights.

FIDUCIARY SCOPE

Nominal payments are covered under the case's default-free bond assumption. Neither ETF, the modelled return nor the 2030 real goal is guaranteed; trustees must authorise the proposal.

QUANTITATIVE AUDIT

Run analysis/run_initial.py from saved raw data. Inspect outputs/max_utility_A4_weights.csv for tickers/weights and max_utility_preference.csv for the A grid; see DATA_RECORD.md and retrieval_manifest.jsonl for provenance. This is a retrospective case.

MINI GLOSSARY

[1] **CAPM return** Annual arithmetic forecast: risk-free rate + beta x market premium; forecast alpha=0.
[2] **Volatility** Monthly simple-return standard deviation x sqrt(12); a risk estimate, not a loss cap.
[3] **Utility (A)** E[R] - 0.5 x A x annual variance; a larger A penalises risk more.
[4] **CAGR** The constant annual compound rate; the goal hurdle includes retained coupon cash.
[5] **Drawdown** Decline in flow-adjusted growth-sleeve unit value from its previous peak.